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MOTIVATE
INNOVATE
EMPOWER

27
YEARS

LIBRARY BOOK TRACKING SYSTEM USING DOUBLY LINKED LIST

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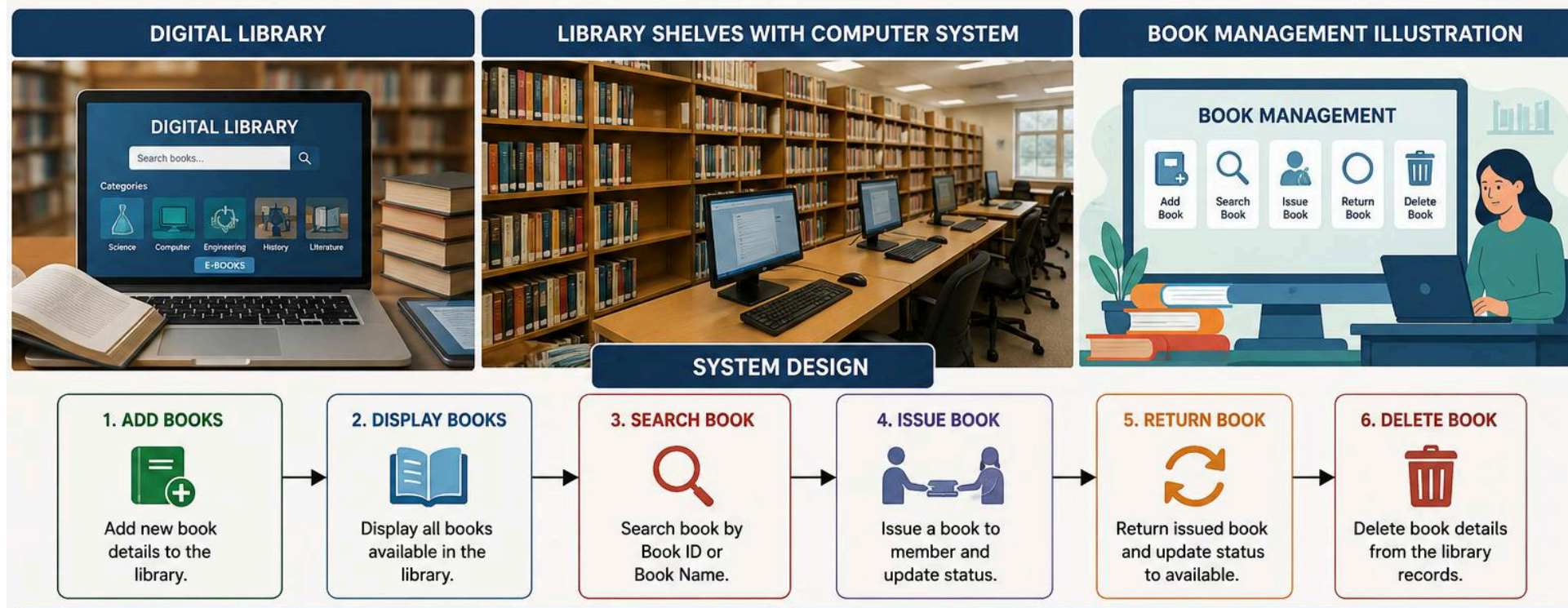


Content

- ✓ Library Book Tracking System is used to manage library records efficiently.
- ✓ It helps to store, search, issue, and return books easily.
- ✓ The system reduces manual work and saves time.
- ✓ Doubly Linked List is used for efficient insertion and deletion operations.

BENEFITS OF DOUBLY LINKED LIST

- Efficient insertion and deletion
- Easy traversal in both directions
- Ideal for dynamic library operations



- ✓ Manual library records are difficult to maintain.
- ✓ Searching books manually consumes more time.
- ✓ Maintaining issue and return records is complicated.
- ✓ The system provides fast and efficient book management.
- ✓ The project reduces paperwork and improves accuracy.

Student using Digital Library system

-  **Search Books**
Quickly search books by title, author or category.
-  **Issue / Return Books**
Issue and return books with just a few clicks.
-  **Track History**
View issued books and history in your account.
-  **Easy & Convenient**
Access the library anytime, anywhere.

MANUAL vs DIGITAL LIBRARY SYSTEM

MANUAL LIBRARY SYSTEM



- ✗ Time consuming process
- ✗ Hard to search and track books
- ✗ Manual record maintenance
- ✗ More chances of errors and loss
- ✗ Limited accessibility
- ✗ Requires more physical space

VS

DIGITAL LIBRARY SYSTEM



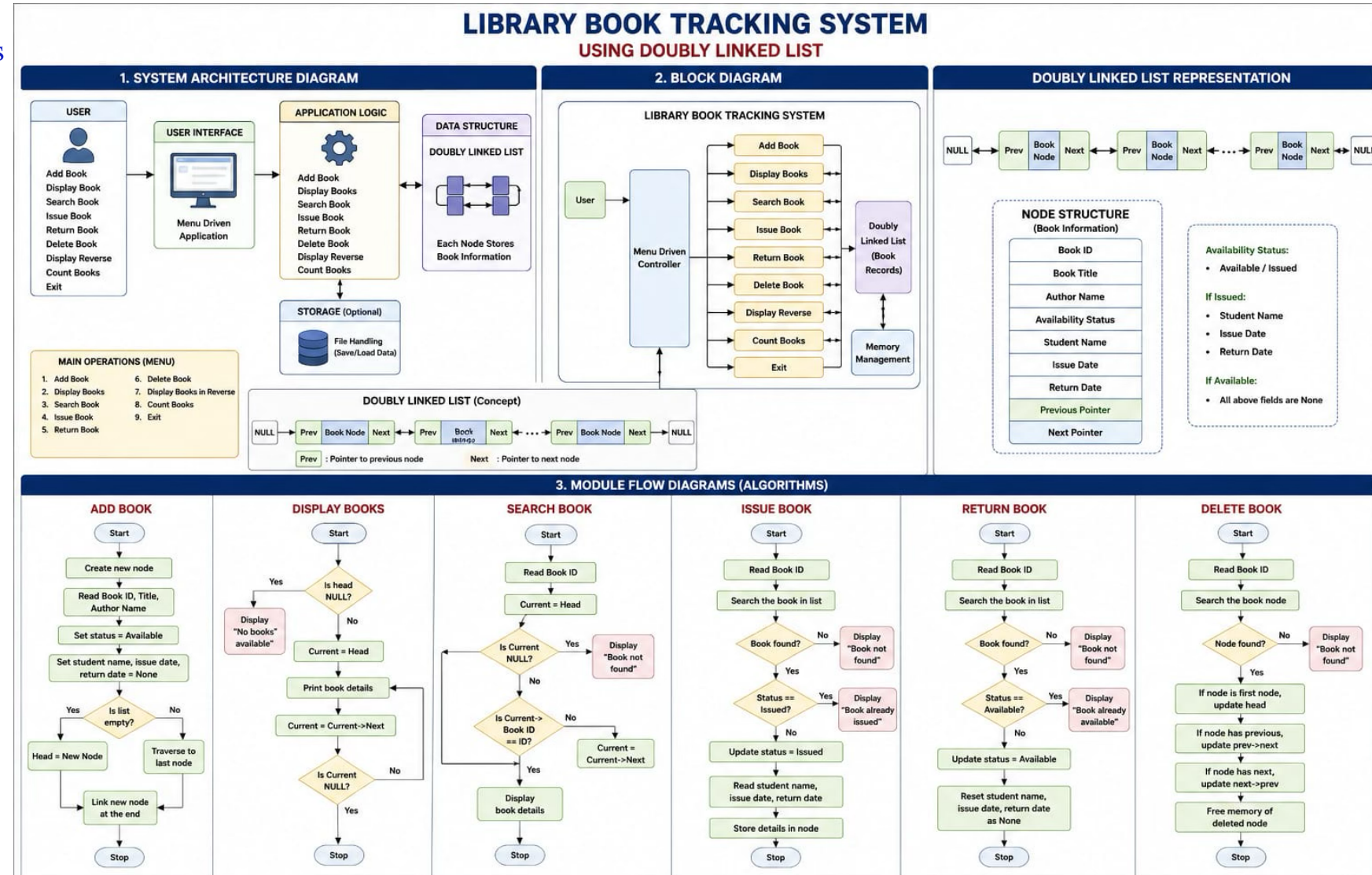
- ✓ Fast and efficient process
- ✓ Easy search and tracking
- ✓ Digital record maintenance
- ✓ Reduces errors and data loss
- ✓ Accessible anytime anywhere
- ✓ Saves physical space

Problem Statement

- ✓ Traditional library systems consume time
- ✓ Difficulty in maintaining issue/return records
- ✓ Need for dynamic memory management
- ✓ Requirement of efficient data organisation

System Design

- ✓ Add Book Module
- ✓ Display Book Module
- ✓ Search Book Module
- ✓ Issue Book Module
- ✓ Return Book Module
- ✓ Delete Book Module



Doubly Linked List

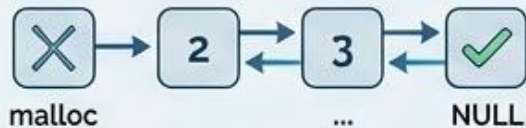
- ✓ Each node stores book information.
- ✓ Every node contains previous and next pointers.
- ✓ Supports forward and backward traversal.
- ✓ Dynamic memory allocation is used.
- ✓ Insertion and deletion operations are efficient.

ALGORITHM OVERVIEW



Add Book

(malloc(), traverse to end)



Delete Book

(handle *prev, *next pointers)

→ handle *prev

← *next pointers

→ free memory

Library Book Tracking System

Based on Doubly Linked List (DLL)

USER COMMAND MENU

[+] Add Book

[i] Display Books

[?] Search Book

[→] Issue Book

[←] Return Book

[x] Delete Book

[↔] Display in Reverse

[#] Count Books

[x] Exit

Node Structure Design

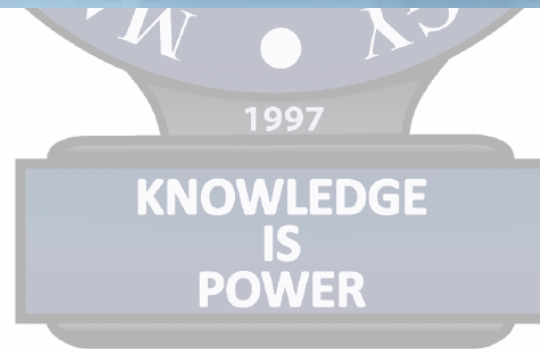
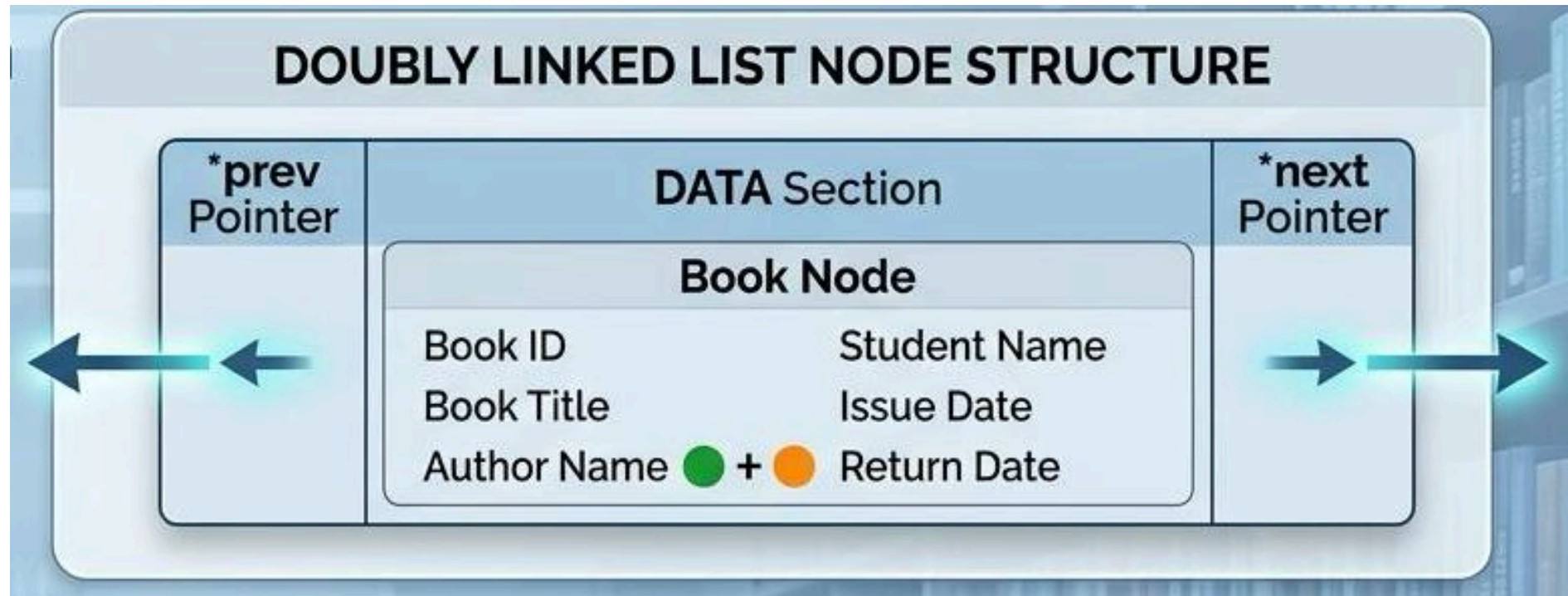
```
struct Book {  
    int bookID;  
    char title[50];  
    char author[50];  
    char status[20];  
    struct Book *prev;  
    struct Book *next;  
};
```

Explanation

Stores complete book details

prev points to previous node

next points to next node



Program Implementation

✓ Programming Language Used: C Language

✓ Concepts Used:

Structures

Functions

Pointers

Dynamic Memory Allocation

Doubly Linked List

✓ The program performs all library operations efficiently.

Features of the System

✓ book management

✓ Dynamic insertion and deletion

✓ Efficient searching

✓ User-friendly operations

✓ Maintains issue and return records



Advantages & Limitations

Advantages

Efficient memory usage

Fast operations

Easy traversal

Limitations

Console-based interface

No database connectivity

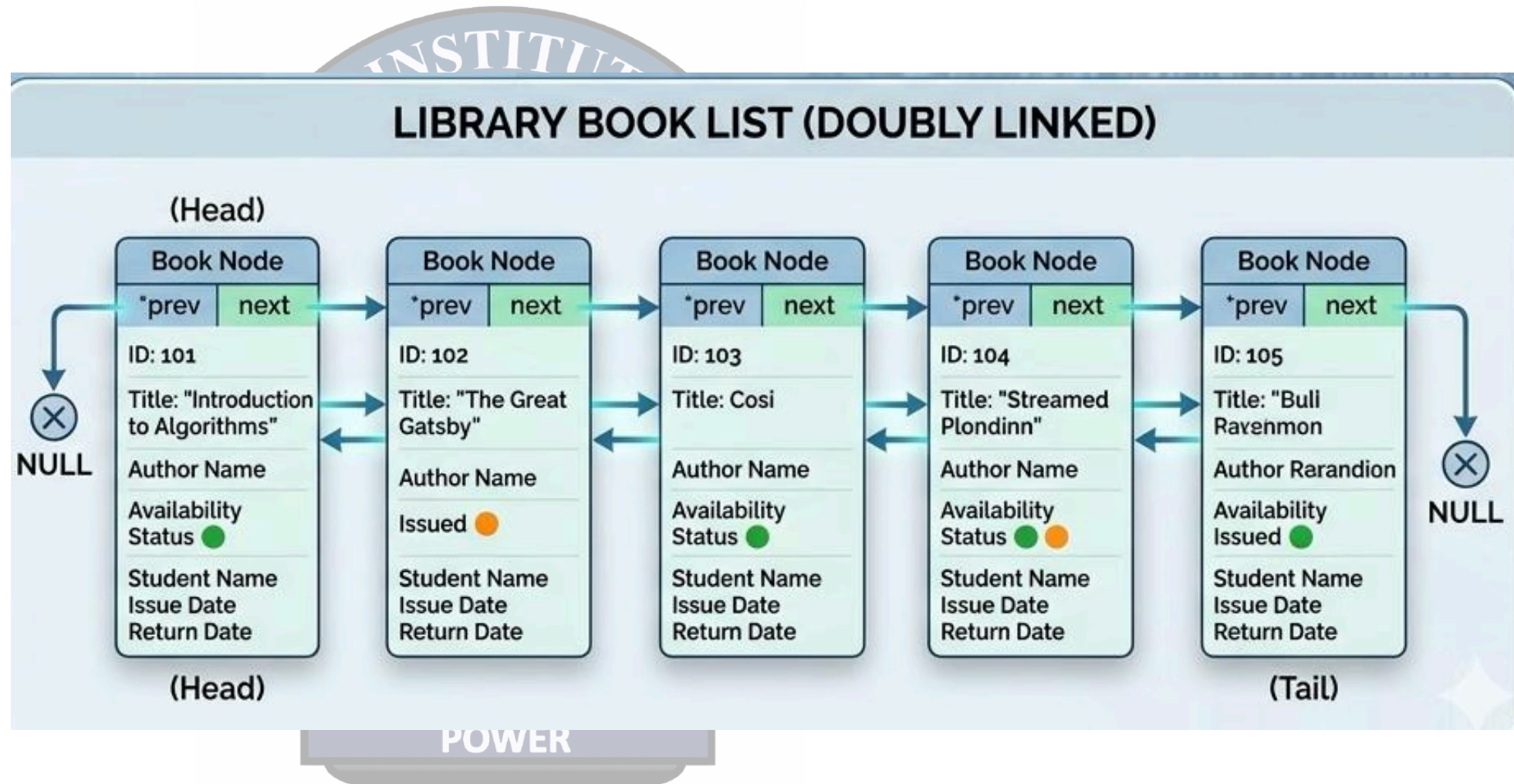
Applications

School Libraries

College Libraries

Personal Book Management

Digital Record Systems



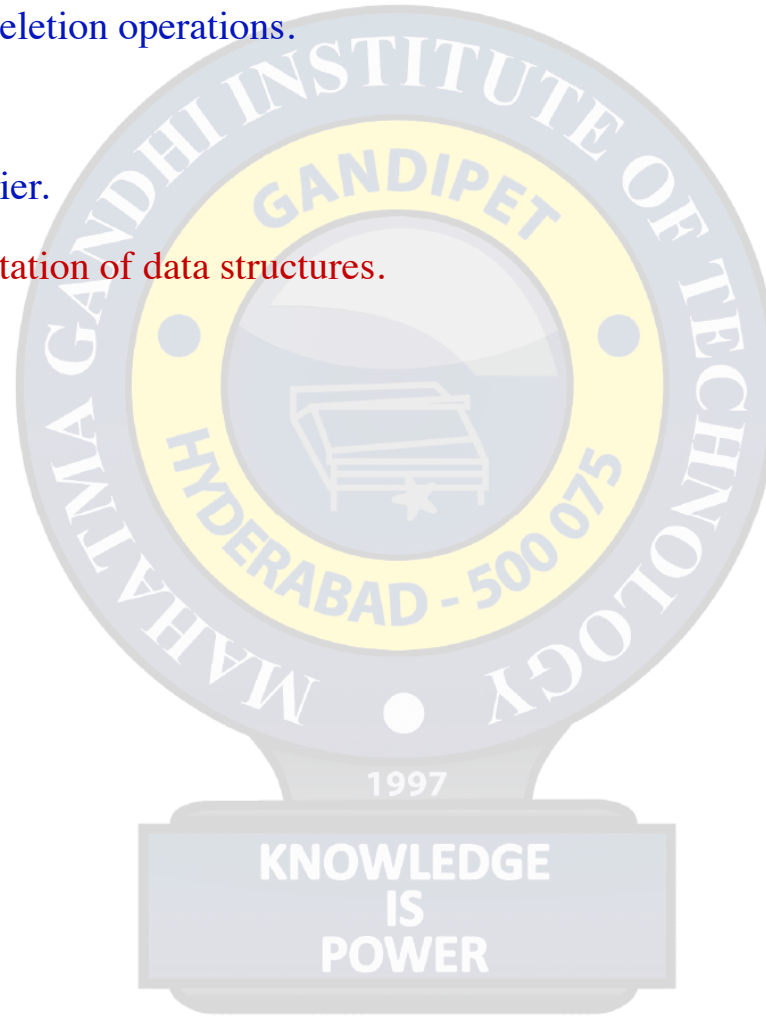
Conclusion

- ✓ The project efficiently manages library records.
- ✓ Doubly Linked List improves insertion and deletion operations.
- ✓ The system reduces manual effort.
- ✓ Book issue and return operations become easier.
- ✓ The project demonstrates practical implementation of data structures.

Reference

Online Learning Resources

Project Documentation



THANK YOU

